

# Pre-Thesis Brief:

## Architecture Patterns for Real-Time & Safety-Critical Cyber-Physical Systems

May 2, 2025

### 1 Goal in One Sentence

Survey, classify and clearly explain the key *architecture patterns* that engineers rely on to guarantee **real-time determinism** and **safety** in cyber-physical systems (CPS), then identify the open gaps—all inside a polished, ~15-page pre-thesis document.

### 2 Concrete Deliverables

1. **Scope paragraph** ( $\frac{1}{2}$  page) — what is in / out of scope.
2. **Mini systematic mapping** — search IEEE Xplore, ACM DL and Scopus, keep  $\approx 20$  core papers; show a spreadsheet or Zotero library.
3. **Taxonomy diagram** — 2–3 axes to organise the patterns (e.g. Timing basis, Fault-tolerance style, Partitioning scope).
4. **Pattern mini-chapters** (6–7,  $\frac{1}{2}$  page each), each containing:
  - Intent sentence
  - Structure diagram
  - Timing or fault scenario sketch
  - Short pros/cons table
  - Reference to at least one real product or standard
  - Link back to a clause in a safety norm (ISO 26262, IEC 61508, DO-178C, ...)
5. **Two case snapshots** ( $\frac{3}{4}$  page each) showing patterns in the wild.
6. **Gap & trend analysis** (1 page) — what is still unsolved and why.
7. **Professional packaging** — 15 pages PDF, consistent figure style,  $\approx 25$  references,  $\leq 5\%$  similarity on plagiarism check.

### 3 Suggested Section Layout

1. Introduction (motivation, definitions)
2. Methodology (literature search and filtering)
3. Taxonomy
4. Core patterns (TTA, Separation Kernel, N-Modular Redundancy, DDS/Publisher-Subscriber, AUTOSAR Classic vs. Adaptive, Watchdog/Safety-Supervisor, TSN/TTE)
5. Integrated safety-security patterns

6. Case snapshots (e.g. Airbus A350 network, automotive drive-by-wire)
7. Gap analysis & future trends
8. Conclusion

## 4 12-Week Work Plan

| Week | Hand-in                                       |
|------|-----------------------------------------------|
| 1    | Scope paragraph                               |
| 2–3  | Literature list & inclusion criteria          |
| 4    | One-page annotated outline + taxonomy draft   |
| 5–7  | Pattern write-ups & diagrams                  |
| 8    | Case snapshots                                |
| 9    | Draft Sections 5–6; internal plagiarism check |
| 10   | Gap analysis page                             |
| 11   | Full draft to supervisor; integrate feedback  |
| 12   | Final polish & PDF submission                 |

## 5 Template for *One* Pattern Subsection

1. **Intent** (problem statement)
2. **Context / forces**
3. **Structure diagram**
4. **Timing or fault scenario**
5. **Safety argument** (link to standard clause)
6. **Known uses**
7. **Trade-offs**

## 6 Starter Reading Pack

- H. Kopetz, *Real-Time Systems* — Time-Triggered basics
- J. Lala & R. Harper, “Architectural Principles for Safety-Critical Real-Time Applications”
- S. Frey *et al.*, “Safe and Secure System Architectures for CPS”
- ISO 26262-6, IEC 61508-3, DO-178C (quote exact clauses)

## 7 Path to the Bachelor Thesis

Choose *one* unresolved gap and turn it into an empirical or tool-building project, e.g.:

- Implement a time-triggered micro-network on STM32 and measure jitter.
- Compare TSN vs. TTEthernet latency under fault injection in OMNeT++.
- Formally verify an ARINC 653 schedule with UPPAAL.

## 8 Tips for Extra Marks

- Keep diagrams uniform and readable at 100 % zoom.
- Quantify whenever possible (jitter, bandwidth, WCET).
- Cite standards sparingly but precisely (clause numbers).
- End each section with one honest limitation.